13.2 ALTERNATING CURRENT (A.C.) CIRCUIT THEORY

(*i*) A.C. Circuit Containing Resistance only

Suppose an alternating sinusoidal voltage

$$E = E_0 e^{j\omega t}$$

is applied across a pure resistance R (Fig. 13.4).

The current through the resistance at any instant t is,

$$I = \frac{E}{R} = \frac{E_0 e^{j\omega t}}{R} = I_0 e^{j\omega t}$$

Here, $$I_0 = \frac{E_0}{R}.$$

Actual value of current = Imaginary part of $I_0\, e^{j\omega t} = I_0 \sin \omega t$.

$\therefore$ $$I = I_0 \sin \omega t$$

Hence the current is in phase with the driving voltage (Fig. 13.5).

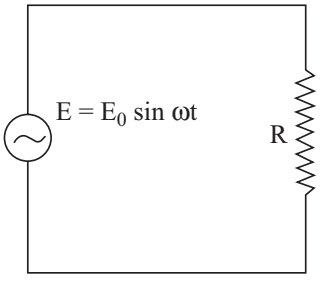


FIG. 13.4

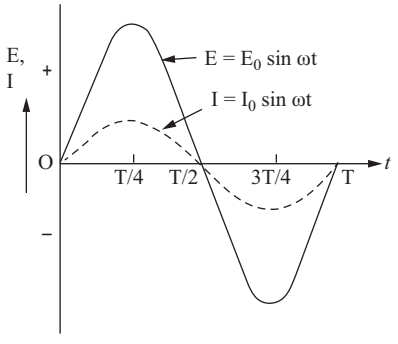


FIG. 13.5

(*ii*) A.C. Circuit Containing Inductance only

Let an alternating voltage

$$E = E_0\, e^{j\omega t}$$

be applied across an inductance L (Fig. 13.6). Let I be the instantaneous current and dI/dt be the rate of change of current with time through the inductance.

The voltage induced across the inductance $= L\dfrac{dI}{dt}$.

$$L\frac{dI}{dt} = E_0\, e^{j\omega t}$$

or $$dI = \frac{E_0}{L} e^{j\omega t}\, dt$$

Integrating, $$I = \frac{E_0}{j\omega L} e^{j\omega t} = \frac{E}{j\omega L} \quad \ldots(1)$$

$\therefore$ $$I = \frac{E_0}{\omega L} e^{j(\omega t - \pi/2)}$$

Let $I_0 = \dfrac{E_0}{\omega L}$ be the maximum current. Then,

$$I = I_0 e^{j(\omega t - \pi/2)}$$

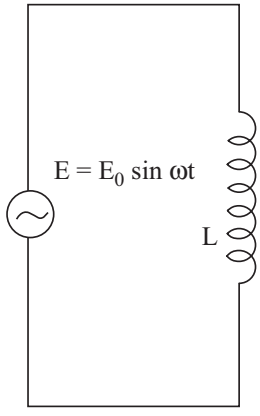


FIG. 13.6

The actual current in the circuit is

$$I = I_0 \sin\left(\omega t - \frac{\pi}{2}\right) \qquad ...(2)$$

In a *pure inductor*, *the current lags behind the emf by* $\pi/2$ (Fig. 13.7).

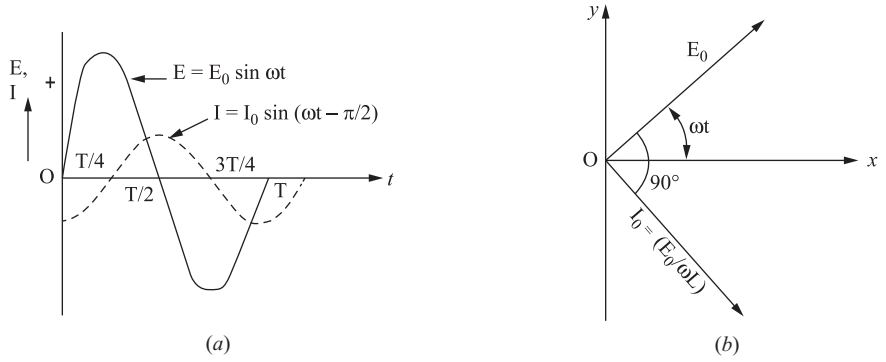


FIG. 13.7

(*iii*) A.C. Circuit Containing Capacitance only

Let an alternating voltage,

$$E = E_0 e^{j\omega t}$$

be applied across the plates of a capacitor of capacitance C (Fig. 13.8). Let q be the charge at any instant across the plates of the capacitor. The p.d. across the plates at any instant must be equal to the applied emf at that instant. Thus

$$\frac{q}{C} = E_0 e^{j\omega t}$$

The current through the circuit is given by,

$$I = \frac{dq}{dt} = \frac{d}{dt}(CE_0\, e^{j\omega t}) = j\omega C\, E_0 e^{j\omega t}$$

or

$$I = \frac{E_0}{1/\omega C} e^{j(\omega t + \pi/2)} = I_0 e^{j(\omega t + \pi/2)}$$

Here, $I_0 = \dfrac{E_0}{1/\omega C}$ is the peak value of alternating current.

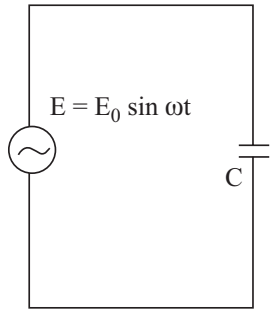


FIG. 13.8

The actual current in the circuit is

$$I = I_0 \sin\left(\omega t + \frac{\pi}{2}\right).$$

In a pure capacitor, *the current leads the emf by* $\pi/2$ (Fig. 13.9)

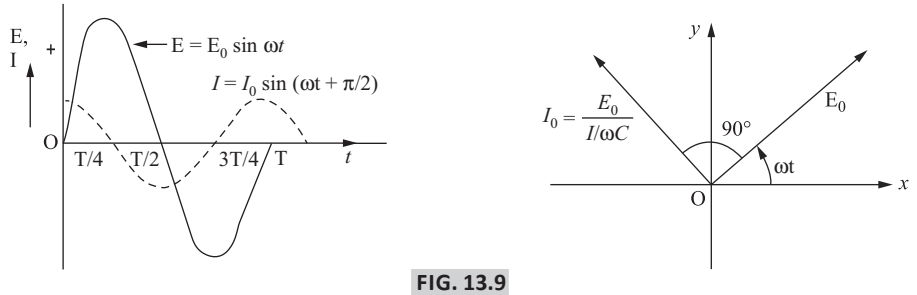


FIG. 13.9